

Per-Particle Spring-Energy Clustering Is Kernel Clustering:

Known Equivalences Behind a Null Screening

Luciano Silva Alarco

Pontificia Universidad Católica del Perú

ORCID 0000-0002-3395-3129

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Abstract

A research line screened clustering objectives built from springs, tension and lifting, and found no mechanical component surviving matched kernel controls. Without its code, we reconstruct the family as *pairwise* spring energies (a pair potential $\phi(d)$, optionally lifted, per particle or summed) and the control as kernel k -means on the same matrix. Per particle, the null result follows from the literature: the energy is Hofmann and Buhmann’s pairwise-clustering cost, hence (up to a constant and a factor $\frac{1}{2}$) a kernel k -means objective, positive semidefinite after a shift (Roth et al.; Dhillon et al.); the kernel is dataset-independent for potentials of negative type (Sejdinovic et al.; França et al.); zero-temperature relaxation is Hartigan’s method in kernel space. The summed energy, equivalently the released tension, is min-sum k -clustering; for non-negative springs A_{ij} it is not a weighted kernel k -means objective, even with data-dependent weights, and stays at uniform distance at least $\frac{1}{2} \min_{i \neq j} A_{ij}$ from all of them. Smaller additions: how a diagonal shift freezes Lloyd’s rule; and, for values rather than minimisers, the converse of the shift invariance (the rest-length spring has no dataset-independent positive definite representative), the uniqueness of the size weight $1/n_c$, and exact certificates that three functionals are not pairwise. Experiments confirm the identities (to 1.1×10^{-11} ; identical spring and kernel trajectories in 90/90 runs). Clustering performance is not addressed.

1 Introduction

Physical metaphors recur in clustering: gravity [33], spin models at finite temperature [3], deterministic annealing [21, 22], quantum potentials [16], elastic nets [9, 34]. The research line LMI001 (“Mechanical analogies for data clustering”) studied a more elementary kind: points connected by springs, with tension and lifting terms, clustered by minimising an elastic energy; its screening found no mechanical effect distinguishable from matched kernel controls. *The definitions of Section 2 are a reconstruction from the line’s public summary (no manuscript or code of the screening was available to us), and so is our reading of “matched kernel control” as kernel k -means on the same matrix with the same optimiser; the conclusions apply to objectives of that pairwise form.* The note explains the null result, and the explanation is not new: it is contained in three literatures that the line’s summary does not mention. Pairwise data clustering [15] minimises exactly the per-particle spring energy of a dissimilarity matrix (by deterministic annealing); its reduction to k -means after a constant shift is the constant-shift embedding of Roth et al. [23]; the same objective is the ratio association that weighted kernel k -means optimises [7, 8]. For potentials of negative type, energy statistics supplies a dataset-independent kernel [29], and clustering by within-cluster energy is kernel k -means, best optimised by Hartigan’s method [10]. A control that matches the kernel therefore matches the per-particle objective exactly, and no experiment on objective values can separate them (experiments on optimisers can, Section 5).

The summed energy, and with it the tension formulation (Theorem 3.4(d)), is min-sum clustering, which for non-negative springs is not a weighted kernel k -means objective (Proposition 4.3).

Contribution. The note collects these results for springs, with explicit constants, and draws the consequence for the research line (Section 3). Its own additions are modest: the mechanism by which a diagonal shift freezes Lloyd’s rule while the energy and the single-move relaxation are unchanged (Proposition 3.10, quantified in E3); the converse of the shift invariance of Roth et al. [23] (Proposition 3.2), whence the rest-length spring, not of negative type, has no dataset-independent PD representative (Corollary 3.6); the size weights that give weighted kernel k -means objectives (only $1/n_c$, Proposition 4.2) and the distance of the total energy from all of them (Proposition 4.3); exact certificates that three functionals are not pairwise under either normalisation (Theorem 4.1; values, not minimisers); a reformulation of the dynamic open problem (Problem 4.4); and a computational confirmation with predefined criteria (Section 5). The new mathematical content is small; the note’s value is to make the null screening of per-particle energies a corollary of known results.

Scope. The note is about objectives and their minimisers, not about clustering quality; we do not claim that any potential clusters real data well. The datasets are standard synthetic ones and two small scikit-learn datasets [20]. Every number in the text is generated by the scripts described in Appendix A, with fixed seeds. Section 6 lists all claims with their status.

2 The pairwise spring family

Throughout, $X = (x_1, \dots, x_n)$ is a configuration in $\mathbb{R}^d$, $k \geq 2$ is fixed, and a *partition* $C = (C_1, \dots, C_k)$ is a partition of $[n]$ into k non-empty clusters, with sizes $n_c = |C_c|$ and indicator vectors $\mathbf{1}_c \in \{0, 1\}^n$. We write $D_{ij} = \|x_i - x_j\|$. “PSD” always refers to a matrix (positive semidefinite); a *kernel* on a set is called *positive definite* (PD) when all its Gram matrices are PSD, the convention of Berg et al. [2].

Definition 2.1 (Spring network, elastic energy, variants). A *spring network* is a triple (ϕ, L, w) of a *pair potential* $\phi: [0, \infty) \rightarrow \mathbb{R}$, a *lift* $L: \mathbb{R}^d \rightarrow \mathbb{R}^D$ (the identity by default), and a *normalisation* $w \in \{w_{\text{sp}}, w_{\text{tot}}\}$ with $w_{\text{sp}}(m) = 1/m$ (energy per particle of the cluster) and $w_{\text{tot}}(m) = 1$ (total energy). Its *within-cluster elastic energy* is

$$E_{\phi, L, w}(C) = \sum_{c=1}^k w(n_c) \sum_{\substack{i < j \\ i, j \in C_c}} \phi(\|Lx_i - Lx_j\|). \quad (1)$$

We write $A_{ij} = \phi(\|Lx_i - Lx_j\|)$ for the *spring matrix*, E_{sp} and E_{tot} for the two normalisations, and $\text{SSE}(C) = \sum_c \sum_{i \in C_c} \|x_i - \mu_c\|^2$, μ_c the centroid, for the k -means objective [18, 19]. The *released tension* of C is the energy of the springs that cross clusters, $T(C) = \sum_{c < c'} \sum_{i \in C_c, j \in C_{c'}} A_{ij}$; a tension formulation maximises T . An *anchored* network attaches every particle to a hub $y_c \in \mathbb{R}^D$ of its cluster, with energy $E^{\text{anch}}(C, y) = \sum_c \sum_{i \in C_c} \phi(\|Lx_i - y_c\|)$, minimised over the hubs as well as over C .

The potentials used below are: the Hooke spring with zero rest length, $\phi(d) = d^2$ (the spring constant is absorbed); the Hooke spring with rest length $r > 0$, $\phi(d) = (d - r)^2$; the string under constant tension, $\phi(d) = d$, whose per-particle energy is the within-cluster dispersion of energy statistics [10]; the saturating (breakable) spring, $\phi(d) = 1 - e^{-d^2/2\ell^2}$; the soft spring, $\phi(d) = \log(1 + d^2/\ell^2)$; and the Hooke spring on the paraboloid lift $L(x) = (x, \|x\|^2/s)$.

Definition 2.2 (Kernel k -means objective, trace form). For a symmetric matrix $K \in \mathbb{R}^{n \times n}$ and a partition C ,

$$J_K(C) = \text{tr } K - \sum_{c=1}^k \frac{\mathbf{1}_c^\top K \mathbf{1}_c}{n_c}. \quad (2)$$

When K is PSD with a Gram factorisation $K_{ij} = \langle \Phi_i, \Phi_j \rangle$, $J_K(C) = \sum_c \sum_{i \in C_c} \|\Phi_i - m_c\|^2$ with m_c the centroid of the Φ_i , $i \in C_c$: the k -means objective in feature space [6, 12, 28, 35]. For general symmetric K we take (2) as the definition.

Definition 2.3 (Negative type, Bernstein functions). A symmetric kernel ψ on a set S is *conditionally negative definite* (CND) if $\sum_{i,j} c_i c_j \psi(s_i, s_j) \leq 0$ for all finite families $s_i \in S$ and reals c_i with $\sum_i c_i = 0$. A function $\phi: [0, \infty) \rightarrow \mathbb{R}$ is of *negative type* if $(x, y) \mapsto \phi(\|x - y\|)$ is CND on $\mathbb{R}^d$ for every d . A function $\psi: [0, \infty) \rightarrow [0, \infty)$ is a *Bernstein function* if it is continuous, C^∞ on $(0, \infty)$, and $(-1)^{m-1} \psi^{(m)} \geq 0$ for all $m \geq 1$; equivalently (Lévy–Khintchine representation, 25, Thm. 3.2) $\psi(s) = a + bs + \int_{(0, \infty)} (1 - e^{-ts}) \mu(dt)$ with $a, b \geq 0$ and $\int \min(1, t) \mu(dt) < \infty$.

3 Results

3.1 Invariances of the trace form

Lemma 3.1 (What J does not see). *Let K be symmetric, C a partition into k non-empty clusters, $\sigma, a \in \mathbb{R}$ and $f \in \mathbb{R}^n$. Then*

$$J_{K+\sigma I}(C) = J_K(C) + \sigma(n - k), \quad J_{K+a\mathbf{1}\mathbf{1}^\top}(C) = J_K(C), \quad J_{K+f\mathbf{1}^\top + \mathbf{1}f^\top}(C) = J_K(C).$$

Proof. $\text{tr}(\sigma I) = \sigma n$ and $\mathbf{1}_c^\top \sigma I \mathbf{1}_c / n_c = \sigma$ for each of the k clusters. $\text{tr}(a\mathbf{1}\mathbf{1}^\top) = an$ and $\sum_c an_c^2 / n_c = an$. $\text{tr}(f\mathbf{1}^\top + \mathbf{1}f^\top) = 2 \sum_i f_i$ and $\mathbf{1}_c^\top (f\mathbf{1}^\top + \mathbf{1}f^\top) \mathbf{1}_c / n_c = 2 \sum_{i \in C_c} f_i$, which sums to $2 \sum_i f_i$ over c . $\square$

So none of the three changes the minimisers of J_K . The lemma is elementary and not new: the diagonal shift is the constant-shift embedding of Roth et al. [23] (a constant added to the off-diagonal dissimilarities) and the device of Dhillon et al. [7, 8] for making an indefinite affinity matrix PSD without changing the optimal partitions; the other two invariances connect the spring energy to Schoenberg’s theory of metric embeddings [26, 27] below.

Proposition 3.2 (Lemma 3.1 is complete). *Let $n \geq 3$, $2 \leq k \leq n - 1$ and $\Delta \in \mathbb{R}^{n \times n}$ symmetric. Then $J_\Delta(C)$ takes the same value on all partitions of $[n]$ into k non-empty clusters if and only if $\Delta = \sigma I + g\mathbf{1}^\top + \mathbf{1}g^\top$ for some $\sigma \in \mathbb{R}$, $g \in \mathbb{R}^n$, and then $J_\Delta \equiv \sigma(n - k)$. The pair (σ, g) is unique, so this space $V_{n,k}$ has dimension $n + 1$ and does not depend on k ; it is also the space of Δ with $J_\Delta(C) = \alpha + \beta(n - k)$ for all partitions with any number of clusters (and then $\alpha = 0$, $\beta = \sigma$). For $k \in \{1, n\}$ there is a single partition and every Δ qualifies. Equivalently, for $2 \leq k \leq n - 1$: $J_K - J_{K'}$ is constant on k -partitions iff the induced dissimilarities $D(K)_{ij} = K_{ii} + K_{jj} - 2K_{ij}$ and $D(K')$ differ by a constant off the diagonal, iff $P(K - K')P \in \mathbb{R}P$ with $P = I - \mathbf{1}\mathbf{1}^\top/n$. [proved here (elementary); checked by an independent internal referee (round 3); proof in Appendix B]*

The proof reduces to zero diagonal and exchanges two points between clusters. Since $J_K(C) = \sum_c \frac{1}{n_c} \sum_{i < j \in C_c} D(K)_{ij}$, the pairwise-clustering cost of a dissimilarity matrix [15] determines the off-diagonal dissimilarities up to one additive constant: the converse of the shift invariance of Roth et al. [23]. We did not find it stated in the literature we know (not searched exhaustively).

Corollary 3.3 (All representatives, and the minimal shift). *Let K be symmetric, $n \geq 3$, $2 \leq k \leq n - 1$. The symmetric K' with $J_{K'} - J_K$ constant on k -partitions are exactly $K' = K + \sigma I + g\mathbf{1}^\top + \mathbf{1}g^\top$. Such a K' can be chosen PSD iff $\sigma \geq \sigma_*(K) := \max\{-c^\top K c : c \perp \mathbf{1}, \|c\| = 1\}$; for $\sigma \geq \sigma_*(K)$, $K' = P(K + \sigma I)P$ is one. [proved here (elementary); checked by an independent internal*

referee (round 3)]. For $K = -A$ with zero diagonal and $\sigma_* \geq 0$, σ_* is the minimal off-diagonal shift of the dissimilarities A_{ij} in Roth et al. [23] (for the dissimilarities $D(K) = 2A$ it is $2\sigma_*$), as we recall their construction (not checked against their text).

Proof. The first sentence is Proposition 3.2 applied to $K' - K$. If $K' \succeq 0$ and $c \perp \mathbf{1}$, then $0 \leq c^\top K' c = c^\top K c + \sigma \|c\|^2$, so $\sigma \geq \sigma_*(K)$. Conversely $P(K + \sigma I)P = K + \sigma I + g\mathbf{1}^\top + \mathbf{1}g^\top$ for some g (expand $P = I - \mathbf{1}\mathbf{1}^\top/n$ and absorb $\mathbf{1}\mathbf{1}^\top$ into g), and it is PSD iff $c^\top (K + \sigma I)c \geq 0$ for all $c \perp \mathbf{1}$. $\square$

3.2 Springs are kernels

Theorem 3.4 (Spring energies and kernel objectives; collected from the literature). *Let (ϕ, L, w) be a spring network with spring matrix A and let $K = -A$ (diagonal included, $K_{ii} = -\phi(0)$). For every partition C into k non-empty clusters:*

- (a) $E_{\phi, L, w}(C)$ depends on the data only through A .
- (b) (per-particle normalisation) $E_{\text{sp}}(C) = \frac{1}{2}J_K(C) + \frac{1}{2}(n-k)\phi(0)$. For every $\sigma \geq -\lambda_{\min}(K)$ the matrix $K + \sigma I$ is PSD and $E_{\text{sp}}(C) = \frac{1}{2}J_{K+\sigma I}(C) + \frac{1}{2}(n-k)(\phi(0) - \sigma)$; hence $\arg \min_C E_{\text{sp}} = \arg \min_C J_{K+\sigma I}$, the k -means problem in the feature space of any Gram factorisation of $K + \sigma I$.
- (c) (Hooke) If $\phi(d) = d^2$ and L is the identity, $E_{\text{sp}}(C) = \text{SSE}(C)$ exactly.
- (d) (total normalisation and tension) $E_{\text{tot}}(C) + T(C) = \sum_{i < j} A_{ij}$ is independent of C ; minimising E_{tot} is maximising the released tension, which is the max- k -cut, equivalently min-sum k -clustering, problem with weights A_{ij} [24]. If all clusters have the same size, $E_{\text{tot}} = (n/k)E_{\text{sp}}$.
- (e) (converse) Every kernel k -means objective is a Hooke energy: if K is PSD and $K = \Phi\Phi^\top$, then $J_K(C) = E_{d^2, \Phi, \text{sp}}(C)$, the per-particle Hooke energy of the lifted configuration $(\Phi_1, \dots, \Phi_n)$.
- (f) (universal kernel) If ϕ is of negative type, then

$$\widetilde{K}(x, y) = \phi(\|x\|) + \phi(\|y\|) - \phi(\|x - y\|) - \phi(0)$$

is a PD kernel on $\mathbb{R}^D$ that does not depend on the dataset (given the scales ℓ, r, s that enter ϕ and L), and $E_{\text{sp}}(C) = \frac{1}{2}J_{\widetilde{K}}(C) + \frac{1}{2}(n-k)\phi(0)$ where $\widetilde{K}$ is evaluated on the lifted points; by (e), up to this constant E_{sp} is the per-particle Hooke energy of any Gram factor of $\frac{1}{2}\widetilde{K}$ on those points.

Status. The theorem collects known results, stated for springs because the explicit constants are what E1 tests. (a), (d) elementary (unlike E_{sp} , E_{tot} is not a weighted kernel k -means objective for non-negative springs, Proposition 4.3); (c) classical; (e) the feature-space reading of k -means [12, 35] [classical]. (b) [literature]: with zero diagonal, E_{sp} is the pairwise-clustering cost of Hofmann and Buhmann [15] for the dissimilarities A_{ij} , made PSD by the constant-shift embedding of Roth et al. [23]; equivalently, minimising E_{sp} maximises the ratio association $\sum_c \mathbf{1}_c^\top (-A) \mathbf{1}_c / n_c$, which is kernel k -means with $K = \sigma I - A$ [7, 8]. Specific to springs is only the constant $\frac{1}{2}(n-k)\phi(0)$. (f) [literature]: $\frac{1}{2}\widetilde{K}$ is the distance-induced kernel of Sejdinovic et al. [29] for the semimetric of negative type $\rho(x, y) = \phi(\|x - y\|) - \phi(0)$ with base point 0 ($\rho \geq 0$ by the CND test on two points), $E_{\text{sp}} - \frac{1}{2}(n-k)\phi(0)$ is the within-cluster energy dispersion for ρ , and its equivalence with kernel k -means is in França et al. [10]; we include the short argument.

Proof. (a) is the definition. (b) Since $\sum_{i < j \in C_c} A_{ij} = \frac{1}{2}(\mathbf{1}_c^\top A \mathbf{1}_c - n_c \phi(0))$,

$$E_{\text{sp}}(C) = \frac{1}{2} \sum_c \frac{\mathbf{1}_c^\top A \mathbf{1}_c}{n_c} - \frac{1}{2} k \phi(0), \quad \text{and} \quad \sum_c \frac{\mathbf{1}_c^\top A \mathbf{1}_c}{n_c} = - \sum_c \frac{\mathbf{1}_c^\top K \mathbf{1}_c}{n_c} = J_K(C) - \text{tr } K,$$

with $\text{tr } K = -n\phi(0)$, so $E_{\text{sp}}(C) = \frac{1}{2}J_K(C) + \frac{1}{2}(n-k)\phi(0)$. The shifted form follows from Lemma 3.1, and $K + \sigma I \succeq 0$ for $\sigma \geq -\lambda_{\min}(K)$. (c) For any finite set P with centroid μ , $\sum_{i,j \in P} \|x_i - x_j\|^2 = 2|P| \sum_{i \in P} \|x_i - \mu\|^2$ (expand $\|(x_i - \mu) - (x_j - \mu)\|^2$; the cross terms vanish). Halving and dividing by $|P|$ gives $\frac{1}{n_c} \sum_{i < j \in C_c} d_{ij}^2 = \sum_{i \in C_c} \|x_i - \mu_c\|^2$; sum over c . (d) Every pair $\{i, j\}$ is either within a cluster or across, so $E_{\text{tot}} + T$ is the total spring energy; the equal-size statement is $w_{\text{tot}} = (n/k)w_{\text{sp}}$. (e) By (b) with $\phi(0) = 0$ applied to the configuration Φ : $E_{d^2, \Phi, \text{sp}}(C) = \frac{1}{2}J_{-D_{\Phi}^{(2)}}(C)$, where $(D_{\Phi}^{(2)})_{ij} = K_{ii} + K_{jj} - 2K_{ij}$, so $-D_{\Phi}^{(2)} = 2K - f\mathbf{1}^\top - \mathbf{1}f^\top$ with $f_i = K_{ii}$, and Lemma 3.1 gives $\frac{1}{2}J_{2K}(C) = J_K(C)$. (f) Positive definiteness of $\widetilde{K}$ is the lemma relating CND and PD kernels [2, Ch. 3, §2, Lemma 2.1]: for points $x_1, \dots, x_m$ and reals c_i , apply the CND inequality to the family augmented by $x_0 = 0$, $c_0 = -\sum_{i \geq 1} c_i$; substituting c_0 it reads $\sum_{i,j \geq 1} c_i c_j [\phi(\|x_i - x_j\|) - \phi(\|x_i\|) - \phi(\|x_j\|) + \phi(0)] \leq 0$, i.e. $\sum_{i,j} c_i c_j \widetilde{K}(x_i, x_j) \geq 0$. On the dataset, $\widetilde{K} = K + f\mathbf{1}^\top + \mathbf{1}f^\top - \phi(0)\mathbf{1}\mathbf{1}^\top$ with $f_i = \phi(\|Lx_i\|)$, so $J_{\widetilde{K}} = J_K$ by Lemma 3.1 and the identity follows from (b); the Hooke reading is (e) applied to $\frac{1}{2}\widetilde{K}$. For Hooke springs $\widetilde{K}(x, y) = 2\langle x, y \rangle$. $\square$

Proposition 3.5 (Which springs are of negative type). (i) $\phi(d) = d^2$ is of negative type: for $\sum_i c_i = 0$, $\sum_{i,j} c_i c_j \|x_i - x_j\|^2 = -2\|\sum_i c_i x_i\|^2 \leq 0$. [classical]

(ii) If N is a CND kernel with $N \geq 0$ and ψ is a Bernstein function, then $\psi \circ N$ is CND. [literature]

(iii) Consequently $\phi(d) = \psi(d^2)$ is of negative type for every Bernstein ψ ; this covers $d^{2\alpha}$ for $0 < \alpha \leq 1$ (in particular d), $\log(1 + d^2/\ell^2)$, $1 - e^{-d^2/2\ell^2}$ and $d^2/(1 + d^2/\ell^2)$. The rest-length potential $(d - r)^2$ is not of negative type. [proved here (elementary)]

Proof sketch. (i) Expand $\|x_i - x_j\|^2 = \|x_i\|^2 + \|x_j\|^2 - 2\langle x_i, x_j \rangle$; the first two terms vanish against $\sum c_i = 0$. (ii) is in Berg et al. [2, Ch. 3, §2] (Schoenberg's theorem is Theorem 2.2 there; we do not vouch for the numbering of the composition result), see also Schilling et al. [25]. The argument: by Schoenberg's theorem [27] e^{-tN} is PD for $t > 0$, so $1 - e^{-tN}$ is CND, and $\psi(N) = a + bN + \int (1 - e^{-tN})\mu(dt)$ is a non-negative combination and pointwise limit of CND kernels. (iii) The listed ψ are Bernstein: s^α ($0 < \alpha \leq 1$), $\log(1 + s/\ell^2)$, $1 - e^{-s/2\ell^2}$, $s/(1 + s/\ell^2)$ all have completely monotone derivatives. If ϕ is of negative type, the CND inequality for two points with $c = (1, -1)$ gives $2\phi(0) - 2\phi(d) \leq 0$, i.e. $\phi(d) \geq \phi(0)$ for all d ; for $(d - r)^2$, $\phi(r) = 0 < r^2 = \phi(0)$. *Rest length.* Theorem 3.4(b) still applies to $(d - r)^2$: on each dataset its minimisers are feature-space k -means minimisers after a diagonal shift. What fails is a dataset-independent PD kernel: $\widetilde{K}(x, x) = -2r^2 < 0$ when $\|x\| = r$, and no dataset-independent representative is PD (Corollary 3.6); E1 finds $\widetilde{K}$ indefinite on every dataset. $\square$

Corollary 3.6 (The rest-length spring has no dataset-independent PD representative). Let $\phi: [0, \infty) \rightarrow \mathbb{R}$ and $A[X]_{ij} = \phi(\|x_i - x_j\|)$. Call a symmetric $\kappa: \mathbb{R}^D \times \mathbb{R}^D \rightarrow \mathbb{R}$ a dataset-independent representative of ϕ if for every set $X \subset \mathbb{R}^D$ of $n \geq 3$ distinct points and some $k \in \{2, \dots, n-1\}$ (equivalently, by Proposition 3.2, every such k), $J_{\kappa[X]} - J_{-A[X]}$ is constant on the partitions of X into k clusters.

(a) κ is a dataset-independent representative iff there are $g: \mathbb{R}^D \rightarrow \mathbb{R}$ and $\sigma \in \mathbb{R}$ with $\kappa(x, y) = -\phi(\|x - y\|) + g(x) + g(y)$ for $x \neq y$ and $\kappa(x, x) = -\phi(0) + \sigma + 2g(x)$. This is the class $\widetilde{K}(x, y) + f(x) + f(y) + a + \sigma[x = y]$ (f, a, σ fixed).

(b) For $\phi(d) = (d - r)^2$, $r > 0$, no dataset-independent representative is PD. Quantitatively, on $X_s = \{0, su, 2su, 3su\}$ ($\|u\| = 1$) the minimal shift of Corollary 3.3 satisfies $\sigma_*(-A[X_s]) \geq 2rs$, which is unbounded in s .

(c) (b) still holds if $J_{\kappa[X]}$ is only required to equal $\lambda_X J_{-A[X]}$ plus a constant, with a factor $\lambda_X > 0$ that may depend on X .

[proved here (elementary); checked by an independent internal referee (round 3); proof of (c) in Appendix B]

Proof of (a) and (b). (a) *If:* Lemma 3.1. *Only if:* by Proposition 3.2, for each X there is a unique (σ_X, g_X) with $\kappa[X] + A[X] = \sigma_X I + g_X \mathbf{1}^\top + \mathbf{1} g_X^\top$. If $X \subseteq Y$, the principal submatrix of Y 's decomposition is a decomposition for X , so by uniqueness $\sigma_X = \sigma_Y$ and $g_X = g_Y|_X$. Comparing any X, X' through $X \cup X'$ shows that $\sigma_X = \sigma$ is one constant and that $g(x) := g_X(x)$ does not depend on the set $X \ni x$. (b) For $c = (1, -1, -1, 1)$ on X_s the terms in g vanish against $\sum_i c_i = 0$, and $\sum_{i,j} c_i c_j \phi(d_{ij}) = 4\phi(0) + 2[\phi(3s) - \phi(s) - 2\phi(2s)] = 4r^2 + 2(4rs - 2r^2) = 8rs$, so $\sum_{i,j} c_i c_j \kappa(x_i, x_j) = -8rs + 4\sigma < 0$ for $s > \sigma/2r$ and κ is not PD. Also $c/2$ is a unit vector orthogonal to $\mathbf{1}$ with $(c/2)^\top A[X_s](c/2) = 2rs$, whence $\sigma_* \geq 2rs$. $\square$

Remark 3.7 (Same minimisers instead of same values). (a) If a fixed symmetric Δ satisfies $\arg \min J_{K+\Delta} = \arg \min J_K$ for every positive definite matrix K (in particular, if it does so for every symmetric K ; $n \geq 3$, $2 \leq k \leq n-1$), then $\Delta \in V_{n,k}$: otherwise $K = \lambda I - \Delta$ with λ large is PD, $J_{K+\Delta} = J_{\lambda I}$ is constant, and $\arg \min J_K = \arg \max J_\Delta$ is a proper subset. As uniform transformations of the matrix, “same minimisers” adds nothing to Proposition 3.2 [proved here (one line); checked by an independent internal referee (round 3)]. (b) On a fixed dataset the weaker notion is vacuous: $K + \sigma I$ ($\sigma \geq \sigma_*$) is PSD with the same minimisers, and so is the block matrix $\sum_c \mathbf{1}_c \mathbf{1}_c^\top$ of a unique minimiser. (c) Open: whether a dataset-independent PD kernel κ has $\arg \min J_{\kappa[X]} = \arg \min J_{-A[X]}$ for all X , k for the rest-length spring. Corollary 3.6(c) excludes positive affine reparametrisations; on three points with $k = 2$, $J_K(\{i\}, \{j, l\}) = \frac{1}{2} D(K)_{jl}$, so the pair closest in the feature metric of κ must be the pair whose distance is closest to r , an ordinal condition that this linear algebra does not reach.

Remark 3.8 (Lifting and anchoring). (a) $E_{\phi, L, w}(X) = E_{\phi, \text{id}, w}(LX)$: a lift only replaces A by $\phi(\|Lx_i - Lx_j\|)$; for Hooke springs it is the choice of the kernel $\langle Lx, Ly \rangle$, and every PD kernel arises this way on finite data (Theorem 3.4(e)) [trivial]. (b) For Hooke springs, $\min_y E^{\text{anch}}(C, y) = \text{SSE}(C) = E_{\text{sp}}(C)$ (hubs at the centroids), and alternating equilibration of hubs and attachments is Lloyd’s algorithm [18]; with $\phi(d) = d$ the anchored objective is k -median [classical]. We make no claim about whether such anchored objectives are of the form (1).

3.3 Relaxation of the spring system

The mechanical way to minimise (1) is to let the system relax: one particle at a time jumps to the cluster in which its springs store less energy, or, at finite temperature, jumps with Metropolis probability. On assignments this is a known algorithm.

Proposition 3.9 (Zero-temperature relaxation is Hartigan’s method). *Let K' be any PSD matrix with $J_{K'} = J_K + \text{const}$ on partitions (for instance $K + \sigma I$ or, for ϕ of negative type, $\widehat{K}$; by Corollary 3.3 these are exactly the PSD matrices $K + \sigma I + g \mathbf{1}^\top + \mathbf{1} g^\top$), and write $\delta_{K'}(i, c) = K'_{ii} - \frac{2}{n_c} (K' \mathbf{1}_c)_i + \frac{1}{n_c^2} \mathbf{1}_c^\top K' \mathbf{1}_c$ for the squared feature distance from Φ_i to the centroid of cluster c . Moving particle i from its cluster a ($n_a \geq 2$) to cluster c changes E_{sp} by*

$$\Delta E = \frac{1}{2} \left[\frac{n_c}{n_c+1} \delta_{K'}(i, c) - \frac{n_a}{n_a-1} \delta_{K'}(i, a) \right], \quad (3)$$

and the right-hand side does not depend on the representative K' . Consequently:

- (i) the greedy single-particle relaxation of the spring energy is Hartigan’s method [13, 14] applied to the kernel k -means objective;

(ii) every partition that is stable under single-particle moves is Voronoi-stable in the feature space of every PSD representative K' , i.e. each Φ_i is assigned to a nearest centroid, so it is a fixed point of Lloyd's algorithm for K' with ties broken in favour of the current cluster (on the relation between such fixed points and local optimality see 30);

(iii) the converse fails: $X = \{0, 2, 3.4, 3.6\} \subset \mathbb{R}$, $k = 2$, $C = \{\{0, 2\}, \{3.4, 3.6\}\}$ is Voronoi-stable with $\text{SSE} = 2.02$, while moving the point 2 gives $\text{SSE} = 1.52$.

Proof. Write $S_{ic} = (K'\mathbf{1}_c)_i$ and $Q_c = \mathbf{1}_c^\top K'\mathbf{1}_c$. Removing i from a changes Q_a/n_a to $(Q_a - 2S_{ia} + K'_{ii})/(n_a - 1)$, adding it to c changes Q_c/n_c to $(Q_c + 2S_{ic} + K'_{ii})/(n_c + 1)$, and $\Delta J = -\Delta \sum_c Q_c/n_c$. Expanding both differences and comparing with $\delta_{K'}$ gives $\Delta J = \frac{n_c}{n_c+1}\delta_{K'}(i, c) - \frac{n_a}{n_a-1}\delta_{K'}(i, a)$, an algebraic identity valid for any symmetric K' ; $\Delta E = \frac{1}{2}\Delta J$ by Theorem 3.4(b). Independence of the representative: if $J_{K'} = J_K + \text{const}$ on partitions then $\Delta J_{K'} = \Delta J_K$ for every move, and the identity holds for both matrices, so the right-hand side of (3) is the same for every such K' . (i) is the definition of Hartigan's move. (ii) If no move decreases E then $\frac{n_c}{n_c+1}\delta(i, c) \geq \frac{n_a}{n_a-1}\delta(i, a)$ for all $c \neq a$, and since $\frac{n_c}{n_c+1} < 1 < \frac{n_a}{n_a-1}$ this forces $\delta(i, c) \geq \delta(i, a)$ (not necessarily strictly, whence the tie rule). (iii) Centroids 1 and 3.5; the point 2 is at distance 1 from the first and 1.5 from the second; after the move the clusters $\{0\}$ and $\{2, 3.4, 3.6\}$ (centroid 3) have $\text{SSE} = 0 + 1 + 0.16 + 0.36 = 1.52$; by (3) with $K' = \tilde{K} = 2\langle x, y \rangle$, $\Delta E = \frac{1}{2}(\frac{2}{3} \cdot 2 \cdot 1.5^2 - 2 \cdot 2 \cdot 1^2) = -0.5$. $\square$

Status. (i) [literature]: Hartigan's method in kernel space is the kernel k -groups algorithm of França et al. [10], there derived for energy statistics; the representative-free form (3) is elementary. (ii) is the kernel form of the Euclidean inclusion of Telgarsky and Vattani [31] [proved here]; we could not check whether França et al. [10] state it and claim no priority. (iii) is the standard type of example. The consequence for the research line is that “spring relaxation” cannot converge to anything outside the set of kernel k -means fixed points.

Proposition 3.10 (Lloyd's rule is representation-dependent). *For $K' = K + \sigma I$, $\delta_{K'}(i, c) = \delta_K(i, c) + \sigma(1 + \frac{1-2\cdot[i \in C_c]}{n_c})$: the shift adds $\sigma(1 + 1/n_c)$ to the squared distance from i to every other cluster c and only $\sigma(1 - 1/n_a)$ to the distance to its own cluster a . Hence the Voronoi rule “assign i to $\arg \min_c \delta_{K'}(i, c)$ ” favours staying, and as $\sigma \rightarrow \infty$ every partition becomes Voronoi-stable, while E_{sp} and the single-move dynamics are unchanged. [proved here (two lines); explicit form of an effect noted by Dhillon et al. [7, 8]]*

Proof. $(\sigma I \mathbf{1}_c)_i = \sigma[i \in C_c]$ and $\mathbf{1}_c^\top \sigma I \mathbf{1}_c = \sigma n_c$; substitute in δ . $\square$

Dhillon et al. [7, 8] already discuss how the diagonal shift affects kernel k -means in practice; the proposition is the explicit mechanism. So “kernel k -means with Lloyd's iteration” is one algorithm per PSD representative, while the single-move relaxation is representative-free. The experiments use $\tilde{K}$ of Theorem 3.4(f) ($2\langle x, y \rangle$ for Hooke springs, i.e. ordinary Lloyd) and the naively shifted $-A + \sigma I$ only as a control (E3).

4 What leaves the family

Theorem 3.4(b) makes every per-particle pairwise objective (a sum over within-cluster *pairs* of a fixed function of the data, weighted by $1/n_c$) a kernel k -means objective; the total normalisation, and with it the released tension, is already outside weighted kernel k -means (Proposition 4.3). Within the per-particle family an ingredient can escape only by breaking one of these three features: the sum over pairs (many-body terms), the fixedness of the pair function (parameters that depend on the assignment), or the normalisation. A fourth route replaces the minimisation over partitions by a dynamics on positions. For the first three, specific instances can be certified exactly.

Theorem 4.1 (Exact separation on explicit configurations; computer-assisted). *Let P_1 and P_2 be the configurations of 7 points in $\mathbb{Q}^2$*

$$P_1 = \{(5, 10), (6, 2), (4, 4), (2, 2), (3, 7), (0, 6), (8, 8)\},$$

$$P_2 = \{(0, 2), (7, 4), (8, 7), (5, 0), (2, 8), (2, 3), (4, 10)\},$$

drawn once from a dedicated generator (seed 20260932), and let $k = 2$, so that each has 63 partitions. Call a partition function F w -pairwise if $F(C) = \sum_c w(n_c) \sum_{i < j \in C_c} A_{ij} + a$ for some symmetric A and constant a (22 unknowns). Then, with squared distances $s_{ij} = \|x_i - x_j\|^2$, the following holds on both configurations (in parentheses the relative residual $\|b - \Pi b\|/\|b\|$ on P_1 ; P_2 , where $b \in \mathbb{Q}^{63}$ lists the values of F and Π is the orthogonal projection onto the w -pairwise functions):

partition function	w_{sp} -pairwise?	w_{tot} -pairwise?
$\sum_c \frac{1}{n_c} \sum d_{ij}^2$ (Hooke, per particle)	yes (0; 0)	no (0.017; 0.015)
$\sum_c \sum d_{ij}^2$ (Hooke, total)	no (0.179; 0.179)	yes (0; 0)
$\sum_c \frac{1}{n_c} \sum (d_{ij}^2 - \rho)^2$, fixed ρ	yes (0; 0)	no (0.026; 0.030)
$\sum_c \frac{1}{n_c} \sum (d_{ij}^2 - \rho_c)^2$, $\rho_c = \text{cluster mean}$	no (0.143; 0.157)	no (0.109; 0.122)
$\sum_c \frac{1}{n_c} \sum_{i < j < l} 4 \text{area}(ijl)^2$ (three-body)	no (0.331; 0.328)	no (0.039; 0.037)
$\sum_c \binom{n_c}{2}^{-1} \sum d_{ij}^2$ (per spring)	no (0.185; 0.180)	no (0.125; 0.121)

where ρ is the mean of all s_{ij} , ρ_c the mean of s_{ij} over the pairs of cluster c , and $\text{area}(ijl)$ the area of the triangle $x_i x_j x_l$. Each row is one function F , tested against both families.

Proof. For fixed w the w -pairwise functions form a linear subspace of $\mathbb{R}^{63}$ spanned by the columns of an explicit rational design matrix (rank 21 for w_{sp} , 22 for w_{tot} , on both configurations). Exact least squares over $\mathbb{Q}$ (`fractions.Fraction`, script `experiments/identity_check.py`) decides membership: a zero residual exhibits A and a , a non-zero one certifies that F is not in the span. $\square$

The two positive controls (Hooke and fixed rest length under their own normalisation, total Hooke under w_{tot}) confirm that the test is not blind; the yes/no pattern agrees on the two configurations in 12 of the 12 cells (one configuration suffices logically). The two normalisations are different objectives (each fails under the other’s weight), and the three functionals tested (rows 4–6: the cluster-mean rest length and the three-body term, both with $1/n_c$ inside F , and the per-spring average) are neither w_{sp} - nor w_{tot} -pairwise; total-weight versions of rows 4–5 were not tested. The per-spring average is a pairwise sum with the size weight $\binom{n_c}{2}^{-1}$; the certificate only excludes the weights 1 and $1/n_c$. By Cauchy–Binet applied to the rows $(1, x_i^\top)$, $\sum_{i < j < l \in C} 4 \text{area}(ijl)^2 = |C| \det S_C$ with S_C the scatter matrix of C , so the three-body row is $\sum_c \det S_c$, a quartic in the data. The theorem certifies these three functionals, not the classes “many-body terms” or “assignment-dependent rest lengths”, and the class statements would be false: restricted to partitions whose clusters all have at least three points, $\sum_c w(n_c)/(n_c - 2) \sum_{i < j < l \in C_c} (s_{ij} + s_{jl} + s_{il}) = \sum_c w(n_c) \sum_{i < j \in C_c} s_{ij}$.

Values, not minimisers. The certificate concerns objective values up to an additive constant. It does not show that the minimisers differ from those of every pairwise objective: with A free, every partition is the unique minimiser of some w_{tot} -pairwise objective ($A_{ij} = -1$ within its blocks, 0 otherwise), and a strictly increasing non-affine transform of a pairwise objective in general fails the test while keeping its minimisers. Whether these ingredients can produce an *observable* mechanical effect is therefore a separate question that the theorem does not answer.

Size weights. The normalisation is the third route above. For point weights $\omega \in (0, \infty)^n$ fixed before the partition is chosen and a symmetric K , the *weighted kernel k -means* objective [6, 8] is $J_K^\omega(C) = \sum_i \omega_i K_{ii} - \sum_c \omega(C_c)^{-1} \sum_{i, j \in C_c} \omega_i \omega_j K_{ij}$, $\omega(S) = \sum_{i \in S} \omega_i$, which for $K = \Phi \Phi^\top$

is $\sum_c \sum_{i \in C_c} \omega_i \|\Phi_i - m_c\|^2$ with m_c the ω -weighted centroid [literature]. For a size weight w write $F_A^w(C) = \sum_c w(n_c) \sum_{i < j \in C_c} A_{ij}$; only $w(m)$, $2 \leq m \leq n - k + 1$, enters. $\mathcal{G}_\omega = \{J_K^\omega : K \text{ symmetric}\}$ is a linear space of functions of the partition, of dimension at most $\binom{n}{2}$ (the n -dimensional space $\{g\mathbf{1}^\top + \mathbf{1}g^\top\}$ is in the kernel of $K \mapsto J_K^\omega$), and it contains the constants ($J_K^\omega \equiv n - k$ for $K = \text{diag}(1/\omega_i)$); so “up to a constant and a non-zero factor” adds nothing. And F_A^w is linear in A and depends only on its off-diagonal part.

Proposition 4.2 (Size weights inside weighted kernel k -means, fixed point weights). *Let $n \geq 4$, $3 \leq k \leq n - 1$ and $w(2) \neq 0$. The following are equivalent:*

- (i) *there is $\omega \in (0, \infty)^n$ such that for every symmetric A there is a symmetric K with $F_A^w - J_K^\omega$ constant on k -partitions;*
- (ii) *the same for every Hooke spring matrix $A_{ij} = (x_i - x_j)^2$ of n distinct points $x_i \in \mathbb{R}$;*
- (iii) *$w(m) = 2w(2)/m$ for $m = 2, \dots, n - k + 1$.*

Under (iii) one can take $\omega = \mathbf{1}$ and $K = -w(2)A$ (zero diagonal). For $k = n - 1$ only $w(2)$ enters and (i)–(iii) hold for every w . For $k = 2$ and $n \geq 4$, (i) implies that $h(a) + h(n - a)$ does not depend on $a \in \{1, \dots, n - 1\}$, where $h(m) = w(m)\binom{m}{2}$, provided $w(n - 1) \neq 0$ and $w(2) + w(n - 2) \neq 0$; for $n = 4$ this is again (iii). [proved here; checked by an independent internal referee (round 3); proof in Appendix B]

The proof is a dimension count plus the exchange argument of Proposition 3.2. Taking n large, a size weight that works for all n and k is $w(m) \propto 1/m$ for all $m \geq 2$. The case $k = 2$ with $h(a) + h(n - a)$ constant and $w \not\propto 1/m$ is not settled by the dimension count. Point weights that depend on A , as for normalised cut [6, 8], are a weaker requirement; for the total energy they do not help:

Proposition 4.3 (The total energy is not weighted kernel k -means). *Let $2 \leq k \leq n - 2$ and let A be symmetric with $A_{ij} \geq 0$ for $i \neq j$, not all zero (for instance the spring matrix of distinct points for any ϕ with $\phi(d) > 0$ for $d > 0$).*

- (a) *For no $\omega \in (0, \infty)^n$, possibly depending on A , and no symmetric K is $E_{\text{tot}} - J_K^\omega$ constant on k -partitions. If $3 \leq k \leq n - 2$ or $(n, k) = (4, 2)$, the same holds for F_A^w whenever $w(3) > 0$ and $w(3) \geq w(2)$.*
- (b) *For every $\omega \in (0, \infty)^n$, every symmetric K and every $c \in \mathbb{R}$, some k -partition C has $|E_{\text{tot}}(C) - J_K^\omega(C) - c| \geq \frac{1}{2} \min_{i \neq j} A_{ij}$; under the conditions of (a), the same holds for F_A^w with $\frac{1}{2}w(3) \min_{i \neq j} A_{ij}$. The constant $\frac{1}{2}$ cannot be improved.*

[proved here; (a) checked by an independent internal referee (round 3); (b) suggested by that referee, proof re-derived and checked by linear programming by the author; proofs in Appendix B]

The proof exhibits, for each ω , a signed measure ν on seven partitions (the two-block splits of four groups of points) that annihilates $\mathcal{G}_\omega$ and the constants but pairs positively with E_{tot} ; since $\|\nu\|_1 = 4 \sum_{t < u} \hat{\omega}_t \hat{\omega}_u$, it also gives (b). When A has zero entries the bound can vanish, and the distance can tend to 0 as ω degenerates (one non-zero spring A_{12} , $n = 4$, $k = 2$, $\omega_3, \omega_4 \rightarrow 0$). The condition $k \leq n - 2$ is needed: for $k = n - 1$, $E_{\text{tot}}(C) = A_{pq}$ for the two-point cluster $\{p, q\}$, which is $J_K^\omega(C)$ when $D(K)_{pq} = A_{pq}(\omega_p + \omega_q)/(\omega_p \omega_q)$. For $w = 1/m$ the pairing vanishes at $\omega = \mathbf{1}$, as it must; for $w(3) < w(2)$ (e.g. $1/m^2$, $\binom{m}{2}^{-1}$) it changes sign, and whether data-dependent weights represent such w is open (fixed weights are excluded by Proposition 4.2 for $3 \leq k \leq n - 2$ and $(n, k) = (4, 2)$). Thus max- k -cut and min-sum k -clustering (Theorem 3.4(d)) are, as functions of the partition (not as sets of minimisers, Remark 3.7(b)), not weighted kernel k -means objectives for non-negative springs, unlike ratio association and normalised cut [8]. We know of no statement of Propositions 4.2–4.3 in the literature (not searched exhaustively).

The dynamic route. Several physics-inspired algorithms do not minimise an energy over partitions but move the *positions*: gravitational clustering [33], mean shift [5, 11] (query points ascend a fixed kernel density estimate) and its blurring variant, in which all data points move [4], quantum clustering [16], and the finite-temperature correlations of a Potts model [3]. Their output is a function of the trajectory, and Theorem 3.4 says nothing about it.

Open problem 4.4 (Dynamic mechanical clustering). Let $V: (0, \infty) \rightarrow \mathbb{R}$ be a pair potential, let $z(t)$ solve $\dot{z}_i = -\sum_{j \neq i} V'(\|z_i - z_j\|) \frac{z_i - z_j}{\|z_i - z_j\|}$ with $z(0) = X$, and let $P_{T,\varepsilon}(X)$ be the single-linkage partition of $z(T)$ at threshold ε . Characterise the triples (V, T, ε) for which there exist a fixed function $a: [0, \infty) \rightarrow \mathbb{R}$ and a normalisation w such that, for every X , $P_{T,\varepsilon}(X)$ is the *unique* minimiser of $\sum_c w(n_c) \sum_{i < j \in C_c} a(\|x_i - x_j\|)$ over partitions into the number of clusters it produces.

Remark 4.5 (Why “unique”, and how a counterexample could be certified). Without uniqueness the problem is trivial: for $a \equiv 0$ every partition is a minimiser (for $w = 1/n_c$, also for every constant a). With uniqueness, a single configuration whose pairwise distances are all distinct is never a counterexample: $a = -1$ on the distances inside the blocks of $P_{T,\varepsilon}(X)$ and $a = 0$ on the others makes it the unique minimiser for $w \in \{1, 1/n_c\}$. A counterexample needs repeated distances, within one configuration or across configurations that share distances. For finitely many configurations it is a finite system of strict linear inequalities in the values of a on the common distance set (scaled to $F(C) - F(P) \geq 1$ for $C \neq P$), whose infeasibility a Farkas certificate over $\mathbb{Q}$ certifies exactly. We have not carried this out. [proved here; checked by an independent internal referee (round 3)]

For the Gaussian attraction $V(d) = -e^{-d^2/2h^2}$ the flow is a continuous-time, unnormalised Gaussian *blurring* mean shift [4], not standard mean shift. We expect no pair (a, w) for generic (T, ε) (untested); either way, such a proposal must be compared with (blurring) mean shift, on trajectories, not with k -means.

5 Computational confirmation

Data and potentials. Five datasets: blobs, two moons and two circles from scikit-learn’s generators (300 points each in $\mathbb{R}^2$; `make_blobs` with three centres and std 0.9, $k = 3$; `make_moons` with noise 0.06 and `make_circles` with factor 0.45 and noise 0.05, $k = 2$), a random subset of 300 digits from `load_digits` (64 features scaled to $[0, 1]$, $k = 10$) and the standardised iris data (150 points, 4 features, $k = 3$). The six potentials of Section 2, with scales set from the data before any run: ℓ is the median distance to the seventh nearest neighbour, r the tenth percentile of all pairwise distances, s the median of $\|x\|^2$. In all, $5 \times 6 = 30$ configurations. Seeds: 20260930 (E1, E2), 20260931 (E3) and 20260932 (E4).

E1: identical objective values. *Criterion (predefined)*: for every tested partition and every potential, $|E_{\text{sp}} - (\frac{1}{2}J_{K'} + \text{const})| \leq 10^{-9} \max(1, |E_{\text{sp}}|)$ with the constant of Theorem 3.4, for the three representatives $K' \in \{-A, -A + \sigma I, \widetilde{K}\}$; pass means 100%. The partitions are 200 uniformly random ones per configuration, the outputs of Lloyd and of the single-move relaxation on $\widetilde{K}$ (minimally shifted when indefinite), and the ground-truth labels. *Result*: 19285/19285 checks passed, worst relative error 1.1×10^{-11} (the shifted representative uses σ up to 4.4×10^4 , so its identity involves cancellation); the Hooke energy equals the k -means SSE to 3.9×10^{-16} . On random partitions the ratio $E_{\text{tot}}/E_{\text{sp}}$ ranges over $[29.7, 197.5]$ (and equals n/k to 2.2×10^{-16} on equal sizes): the normalisation is a genuine choice. The spectrum of $\widetilde{K}$ (per configuration in `results/tables_identity.md`): for the 25 configurations with a potential of negative type it is PSD in 25 (numerically, $\lambda_{\min}/\lambda_{\max} \geq -3.2 \times 10^{-16}$), and for the rest-length potential it is indefinite in 5/5.

E2: same optimiser, same initialisation, same trajectory. *Criterion:* a brute-force spring relaxation that recomputes the total energy from scratch for every candidate move (it sees only springs) and the kernel implementation of Hartigan’s method via (3) (it sees only K), started from the same random labels with the same sweep order, must produce identical label vectors after every sweep and final energies equal to 10^{-9} relative, in all runs. Subsets of 120 points, three initialisations per configuration. *Result:* 90/90 identical trajectories (mean 4.9 sweeps, at most 11), worst final-energy discrepancy 1.4×10^{-14} . In addition, hub-anchored Hooke springs equilibrated alternately (Remark 3.8(b)) reproduced scikit-learn’s Lloyd k -means from the same initial centroids in 15/15 runs, with inertia equal to the spring energy to 4.3×10^{-16} .

E3: relaxation versus kernel k -means on the same energy. Five optimisers see the same kernel: *lloyd* (batch Lloyd on $\widehat{K}$, minimally shifted only when indefinite, 5 configurations), *lloyd+relax* (Lloyd, then single-move polish), *relax* (single-particle relaxation), *anneal* (Metropolis on the spring energy, 40 sweeps of geometric cooling, then relaxation [17]; Hofmann and Buhmann [15] anneal the same energy deterministically) and the control *lloyd (naive)* (Lloyd on $-A + \sigma I$, Proposition 3.10), started from 10 random labelings and 10 kernel k -means++ seedings [1] per configuration (annealing: the first 5 random ones). *Criteria (predefined):* (C3a) every relaxation and annealing fixed point is Voronoi-stable for the kernel used (Proposition 3.9(ii) predicts 100%); (C3b) the relaxation is distinguishable from the kernel family only if its best-of-restarts energy differs from that of *lloyd+relax* by more than 10^{-6} relative (either sign: a statement about optimisers of one function); (C3c) the fraction of Lloyd fixed points that are not single-move stable is reported. *Disclosure:* a pilot run used $-A + \sigma I$ as the only Lloyd kernel and found it inert (Proposition 3.10); $\widehat{K}$ replaced it, which favours the kernel side, and the naive shift was kept as a control. Afterwards only C3a’s wording changed (to name the kernel); every number comes from the final code. *Results:* all 600/600 relaxation and 150/150 annealing fixed points are Voronoi-stable (C3a). Under C3b, 6 of 30 configurations are distinguishable (5 in favour of the relaxation, 1 against, largest relative difference 7.4×10^{-3} ; the split is optimiser noise: the **-fast** run gives 3–3). Only 34% of Lloyd fixed points on the canonical kernel are single-move stable, 6% on the naive shift (C3c). The best energy found by any method, E^* , is reached by the relaxation in 28 configurations, by *lloyd+relax* in 25, by Lloyd alone in 14 and by the naive shift in 6. Tables per configuration and per potential (including annealing and the agreement of the best partitions) are in `results/tables_relaxation.md` and `results/tables_e3_by_potential.md`, excess energies in `figures/energies.pdf`. E3 is descriptive (counts over 30 configurations, modest restart budgets); no inferential claim is made.

Reading of E1–E3 for the research line. If the screened formulations were per-particle pairwise energies (Definition 2.1), the screening could not have found a mechanical effect: the spring energies *are* the kernel objectives (E1), the mechanical implementation of the relaxation *is* a kernel implementation (E2), and every relaxed state *is* a kernel k -means fixed point (E3). Differences between optimisers of the same function are visible (Lloyd alone reaches E^* less often than the single-move methods, consistent with França et al. [10]; the naive shift is worse still), and they are the only differences there are. For the total normalisation the identity is with min-sum k -clustering, not a weighted kernel k -means objective (Proposition 4.3): a null result against a kernel control on the same matrix would be an empirical finding, not a corollary. In particular, tension formulations as reconstructed here (maximising the released tension T) are the total normalisation in disguise (Theorem 3.4(d)) and are not covered by this explanation; a per-particle tension $\sum_c T_c/n_c$, with T_c the energy of the springs cut by cluster c , is ratio cut, a kernel k -means objective [8].

claim	status
J is invariant under diagonal shift, constant and separable terms (Lemma 3.1); for $2 \leq k \leq n-1$ these are all its value invariances (Prop. 3.2), which gives all representatives and the minimal PSD shift (Cor. 3.3)	Lemma: elementary (cf. 8, 23); Prop./Cor.: proved here, r3; converse of 23, not found in the literature we know
Per-particle energy = pairwise-clustering cost = $\frac{1}{2}J_{-A} + \frac{1}{2}(n-k)\phi(0)$, PSD after a shift (Thm. 3.4(b))	literature (7, 8, 15, 23); constant made explicit
Total energy / released tension = min-sum k -clustering / max- k -cut (Thm. 3.4(d))	elementary
Hooke energy = SSE; every kernel k -means objective is a lifted Hooke energy (Thm. 3.4(c), (e)); a lift only changes the kernel, anchored Hooke springs are Lloyd (Remark 3.8)	classical / trivial
Negative-type springs have a dataset-independent PD kernel (Thm. 3.4(f)); Bernstein transforms of d^2 are of negative type (Prop. 3.5)	literature (2, 10, 29) / classical
$(d-r)^2$ is not of negative type (Prop. 3.5) and has no dataset-independent PD representative, even up to a data-dependent factor (Cor. 3.6); same minimisers (Remark 3.7)	proved here; Cor. 3.6, Remark 3.7(a): r3; Remark 3.7(c) open
Single-particle relaxation is Hartigan’s method in kernel space; its fixed points are Voronoi-stable for every PSD representative (Prop. 3.9)	(i) literature (10); (ii) proved here, kernel form of 31 (priority not checked)
Lloyd’s rule depends on the PSD representative; large shifts freeze it (Prop. 3.10)	proved here; effect noted by Dhillon et al. [7, 8]
A three-body term, a cluster-mean rest length (per particle) and the per-spring average are not w -pairwise for $w \in \{1, 1/n_c\}$; the normalisations differ (Thm. 4.1)	proved here (exact arithmetic, two configurations; values, not minimisers)
$1/n_c$ is the only size weight giving weighted kernel k -means for all A with fixed point weights ($k \geq 3$, Prop. 4.2); E_{tot} with $A \geq 0$ is not one even with data-dependent weights, and is at uniform distance $\geq \frac{1}{2} \min A_{ij}$ from all of them (Prop. 4.3)	proved here, r3 (the distance bound suggested by the referee); values, not minimisers; $k=2$ partly open
Dynamic mechanical clustering with unique minimisers (Problem 4.4); distinct distances give no counterexample (Remark 4.5)	open, not tested; remark proved here, r3
Clustering quality on real data	not addressed

Table 1: Mathematical claims and their status; Section 5 only measures (“verified” there means reproduced by the scripts of Appendix A with fixed seeds, not proved). “r3”: new in v0.4, proved here and checked step by step by an independent internal referee in round 3, with its own exact computations for small n ; not externally refereed.

6 What can be claimed

Table 1 collects the mathematical claims with their status. Most of what closes the screening is literature.

Limitations. The family of Section 2 is a reconstruction from the line’s public summary; if the original screening used many-body terms, assignment-dependent parameters, a dynamics on positions or the total normalisation (including the released tension), Section 4 applies and Theorem 3.4 does not explain the null result. The theorem concerns objectives, not optimisers, which differ (E3). The normalisation family is $\{1, 1/n_c\}$ (by Proposition 4.2, $1/n_c$ is the only size weight inside weighted kernel k -means with fixed point weights for $k \geq 3$; $k=2$ and data-dependent weights for decreasing $w \neq 1/m$ remain open). The results on invariances and size weights (Propositions 3.2–4.3, Corollaries 3.3–3.6, Remarks 3.7, 4.5) were checked by one independent internal referee, not by external review; the relation of σ_* to the shift of Roth et al. [23], and whether the converse is known, were not checked against full texts. Theorem 4.1 certifies three functionals on two configurations and concerns objective values, not minimisers; it says nothing about classes of terms or about how far from pairwise those energies are on typical data. The attributions to França et al. [10] and to the diagonal-shift discussion of Dhillon et al. [7, 8] were checked against abstracts and bibliographic indices, not full texts. E3 uses one implementation of each optimiser with modest restart budgets; it is not a benchmark. Nothing is claimed about the dynamic route beyond its formulation.

Next steps for the research line. (a) Confirm with the author of the line that the screened formulations were of the pairwise form of Definition 2.1, and with which normalisation of the energy and of the tension; if not pairwise, the emphasis moves to Section 4. (b) Settle Open problem 4.4 for the Gaussian flow, by configurations with repeated distances on which no fixed pair function makes $P_{T,\varepsilon}$ the unique minimiser (a Farkas certificate, Remark 4.5), or by a representation. (c) The open cases of Propositions 4.2–4.3 ($k = 2$ with $h(a) + h(n - a)$ constant; data-dependent weights for decreasing $w \neq 1/m$). (d) A dataset-independent PD kernel with the minimisers of the rest-length spring (Remark 3.7(c)). (e) Check in Roth et al. [23] the relation of their minimal shift to σ_* , and whether Proposition 3.2 is known, including the literature on shifts of pairwise similarities in pairwise clustering. (f) If (a) is confirmed, update the line’s summary to “no mechanical component of the pairwise, per-particle kind can survive”.

A Reproducibility

`experiments/mechanical.py` implements the potentials, energies, kernel forms and optimisers; `experiments/identity_check.py` runs E1, E2 and E4 and `experiments/relaxation.py` runs E3 (both take `-fast`); `experiments/make_numbers.py` converts `results/*.json` into the macros and table bodies used here, so every number in the text is generated, and `results/*.md` hold the full tables. Seeds 20260930 (E1, E2), 20260931 (E3) and 20260932 (E4, own generator); Python 3.11.15, NumPy 2.4.6, SciPy 1.17.1 [32], scikit-learn 1.9.1 [20]; 159 s (E1, E2, E4) and 128 s (E3) on shared cores. Implementation details are in the README. The exact rank checks of Propositions 3.2–4.3 are in `theory/check_invariances.py` (seed 20261003, about 22s, output saved beside it) and an LP check of Proposition 4.3(b) in `experiments/check_bound_prop43.py`; the integrator’s re-check (not in the repository) is summarised in the README; the independent check is the round-3 internal referee report.

B Proofs of the results on invariances and size weights

Proof of Proposition 3.2. If. Lemma 3.1, with value $\sigma(n - k)$ (the matrix $a\mathbf{1}\mathbf{1}^\top$ is the case $g = \frac{a}{2}\mathbf{1}$). *Uniqueness.* If $\sigma I + g\mathbf{1}^\top + \mathbf{1}g^\top = 0$, the off-diagonal entries give $g_i + g_j = 0$ for $i \neq j$; with a third index l , $g_i = \frac{1}{2}[(g_i + g_j) + (g_i + g_l) - (g_j + g_l)] = 0$, and then $\sigma = 0$. *Only if.* Put $f = \frac{1}{2}(\Delta_{11}, \dots, \Delta_{nn})$ and $e = \Delta - f\mathbf{1}^\top - \mathbf{1}f^\top$, which has zero diagonal and, by Lemma 3.1, $J_e = J_\Delta$. For zero diagonal, $J_e(C) = -2\mathcal{E}(C)$ with $\mathcal{E}(C) = \sum_c \frac{1}{n_c} \sum_{i < j \in C_c} e_{ij}$, so $\mathcal{E}$ is constant on k -partitions. We show $e_{pi} = e_{pj}$ for all distinct p, i, j . If $k \geq 3$, the set $R = [n] \setminus \{p, i, j\}$ has $n - 3 \geq k - 2 \geq 1$ elements; fix a partition of R into $k - 2$ non-empty blocks and compare the partitions $\{\{p, i\}, \{j\}, R\text{-blocks}\}$ and $\{\{p, j\}, \{i\}, R\text{-blocks}\}$: their energies differ by $\frac{1}{2}(e_{pi} - e_{pj})$, which must vanish. If $k = 2$, let $U = [n] \setminus \{i, j\} \neq \emptyset$ and $\gamma_x = e_{ix} - e_{jx}$ for $x \in U$. For $P \subseteq U$ and $Q = U \setminus P$, the partitions $\{P \cup \{i\}, Q \cup \{j\}\}$ and $\{P \cup \{j\}, Q \cup \{i\}\}$ have energies differing by $\frac{1}{|P|+1} \sum_{x \in P} \gamma_x - \frac{1}{|Q|+1} \sum_{x \in Q} \gamma_x$. With $P = \emptyset$ this gives $\sum_{x \in U} \gamma_x = 0$; with $P = \{p\}$ it gives $\gamma_p(\frac{1}{2} + \frac{1}{n-2}) = 0$, so $e_{ip} = e_{jp}$. In both cases, any two pairs are linked by pairs sharing an index ($e_{ab} = e_{ad} = e_{cd}$), so $e_{ij} = -\sigma$ for all $i \neq j$ and some σ , i.e. $e = \sigma I - \sigma\mathbf{1}\mathbf{1}^\top$ and $\Delta = \sigma I + g\mathbf{1}^\top + \mathbf{1}g^\top$ with $g = f - \frac{\sigma}{2}\mathbf{1}$. *All k .* The condition for all partitions implies the one for $k = 2$, hence the form; conversely that form gives $\sigma(n - k)$ for every k . *Equivalent forms.* D is linear with kernel $\{g\mathbf{1}^\top + \mathbf{1}g^\top\}$ (if $D(K) = 0$ then $K_{ij} = \frac{1}{2}(K_{ii} + K_{jj})$), and $D(\sigma I) = 2\sigma(\mathbf{1}\mathbf{1}^\top - I)$; likewise $M \mapsto PMP$ has the same kernel on symmetric matrices and $P(\sigma I)P = \sigma P$. $\square$

Proof of Corollary 3.6(c). Now $\kappa[X] = \lambda_X(-A[X]) + \sigma_X I + g_X \mathbf{1}^\top + \mathbf{1}g_X^\top$. For $X \subseteq Y$, restriction gives $(\lambda_X - \lambda_Y)A[X] \in V_{|X|,2}$. The matrix $A[X_s]$ is not in $V_{4,2}$, because $D(A[X_s])_{ij} = 2\phi(0) - 2\phi(d_{ij})$ is not constant off the diagonal ($(s-r)^2, (2s-r)^2, (3s-r)^2$ are never all equal for $s > 0$). Hence $\lambda_{X_s} = \lambda_{X_s \cup X_{s'}} = \lambda_{X_{s'}} =: \lambda$ for all $s, s' > 0$, the same restriction argument gives one σ for all X_s , and κ/λ fails (b) on X_s for large s . $\square$

Proof of Proposition 4.2. (iii) $\Rightarrow$ (i): with $w(m) = \lambda/m$, $F_A^w = \lambda E_{\text{sp}}$ for the zero-diagonal A , and $E_{\text{sp}} = \frac{1}{2}J_{-A}$ by Theorem 3.4(b). (i) $\Rightarrow$ (ii) is trivial. (ii) $\Rightarrow$ (i): the matrices $M(x)_{ij} = (x_i - x_j)^2$ satisfy $M(x+y) - M(x) - M(y) = 2[(x_i - x_j)(y_i - y_j)]_{ij}$, which for $x = e_a$, $y = e_b$ is $-2(E_{ab} + E_{ba})$; so they span all zero-diagonal symmetric matrices, and since M is polynomial, so do the $M(x)$ with distinct coordinates (a non-empty open set). Membership in the linear space $\mathcal{G}_w$ passes to linear combinations. (i) $\Rightarrow$ (iii): under (i), the $\binom{n}{2}$ functions $F_{E_{ab}+E_{ba}}^w$ ($a < b$) and the constant lie in $\mathcal{G}_w$, of dimension $\leq \binom{n}{2}$, so they are linearly dependent: some $e \neq 0$ (zero diagonal, symmetric) has F_e^w constant. The exchange of the proof of Proposition 3.2, with energies weighted by w , gives $w(2)(e_{pi} - e_{pj}) = 0$, so $e \equiv \varepsilon \neq 0$ off the diagonal and $\sum_c h(n_c)$ is constant over all sizes $(n_1, \dots, n_k)$ of k -partitions. Compare the size vectors $(m+1, 1, \rho)$ and $(m, 2, \rho)$ with the same $k-2 \geq 1$ remaining sizes ρ (possible for $1 \leq m \leq n-k$): $h(m+1) - h(m) = h(2) - h(1) = h(2)$, so $h(m) = (m-1)h(2)$, i.e. $w(m) = 2w(2)/m$. For $k=2$ the exchange with $P = \emptyset$ and $P = \{x\}$ gives $w(n-1)\sum_U \gamma = 0$ and $(w(2) + w(n-2))\gamma_x = 0$, hence again $e \equiv \varepsilon$, and $\sum_c h(n_c) = h(a) + h(n-a)$; for $n=4$, $h(1) + h(3) = 2h(2)$ reads $3w(3) = 2w(2)$. $\square$

Proof of Proposition 4.3. Take $a \neq b$ with $A_{ab} > 0$. Split $[n]$ into $k+2$ non-empty groups with a, b in different groups; keep $k-2$ groups $R_1, \dots, R_{k-2}$ as fixed clusters and call the other four $G_1, \dots, G_4$, with $a \in G_1$, $b \in G_2$; for the statements on F_A^w use singletons $G_1 = \{a\}$, $G_2 = \{b\}$, $G_3 = \{c\}$, $G_4 = \{d\}$ (possible exactly when $k \geq 3$ or $n = 4$). For each of the seven partitions $\{X, Y\}$ of $\{1, 2, 3, 4\}$ into two non-empty sets, let C_X be the k -partition with clusters $G_X = \bigcup_{t \in X} G_t$, G_Y and the R 's, and put $\nu(C_X) = \varepsilon(X)\hat{\omega}(X)\hat{\omega}(Y)$, where $\hat{\omega}(X) = \omega(G_X)$, $\varepsilon(X) = +1$ if $|X| \in \{1, 3\}$ and $\varepsilon(X) = -1$ if $|X| = 2$; $\nu = 0$ on all other partitions. (1) $\sum_X \nu(C_X) = (\hat{\Omega}^2 - \sum_t \hat{\omega}_t^2) - 2\sum_{t < u} \hat{\omega}_t \hat{\omega}_u = 0$, with $\hat{\Omega} = \sum_t \hat{\omega}_t$, because each pair $t < u$ is separated by exactly two of the three 2+2 splits. (2) ν annihilates $\mathcal{G}_w$: by (1) the terms of $J_K^w(C_X)$ that are common to the seven partitions drop out, and what remains is $-\sum_{S \in \{X, Y\}} \hat{Q}(S)/\hat{\omega}(S)$ with $\hat{Q}(S) = \sum_{t, u \in S} \hat{B}_{tu}$, $\hat{B}_{tu} = \sum_{i \in G_t, j \in G_u} \omega_i \omega_j K_{ij}$. Every non-empty proper $S \subset \{1, 2, 3, 4\}$ is a block of exactly one C_X , and $\nu(C_X)/\hat{\omega}(S) = \varepsilon(S)\hat{\omega}(S^c)$; so it suffices that $\sum_{S \ni t, u} \varepsilon(S)\hat{\omega}(S^c) = 0$ for all t, u . For $t = u$: $(\hat{\Omega} - \hat{\omega}_t) - \sum_{v \neq t} (\hat{\Omega} - \hat{\omega}_t - \hat{\omega}_v) + \sum_{v \neq t} \hat{\omega}_v = 0$; for $t \neq u$: $-\hat{\omega}(\{t, u\}^c) + \hat{\omega}(\{t, u\}^c) = 0$. (3) For E_{tot} , $E_{\text{tot}}(C_X) = \text{const} + \sum_{t < u, t, u \text{ on the same side}} \hat{A}_{tu}$ with $\hat{A}_{tu} = \sum_{i \in G_t, j \in G_u} A_{ij} \geq 0$, and $\hat{A}_{12} \geq A_{ab} > 0$. The pair $\{t, u\}$ lies in a two-group block once $(-\hat{\omega}(\{t, u\})\hat{\omega}(\{v, z\}))$ and in a three-group block twice $(\hat{\omega}_v(\hat{\Omega} - \hat{\omega}_v) + \hat{\omega}_z(\hat{\Omega} - \hat{\omega}_z))$, where $\{v, z\}$ is the complementary pair; hence $\langle \nu, E_{\text{tot}} \rangle = 2\sum_{t < u} \hat{A}_{tu} \hat{\omega}_v \hat{\omega}_z \geq 2A_{ab} \hat{\omega}_3 \hat{\omega}_4 > 0$. Since $\langle \nu, J_K^w + \text{const} \rangle = 0$, E_{tot} is not of that form. With singleton groups and a size weight, the same count gives $\langle \nu, F_A^w \rangle = \sum_{t < u} A_{tu} [(w(3) - w(2))\hat{\omega}(\{t, u\})\hat{\omega}(\{v, z\}) + 2w(3)\hat{\omega}_v \hat{\omega}_z] > 0$ under the stated conditions on w . (4) For (b), $g = J_K^w + c$ has $\langle \nu, E_{\text{tot}} - g \rangle = \langle \nu, E_{\text{tot}} \rangle$, so $\max_C |E_{\text{tot}}(C) - g(C)| \geq \langle \nu, E_{\text{tot}} \rangle / \|\nu\|_1$. With $\hat{e}_2 = \sum_{t < u} \hat{\omega}_t \hat{\omega}_u$, $\|\nu\|_1 = \sum_t \hat{\omega}_t (\hat{\Omega} - \hat{\omega}_t) + \sum_{2+2} \hat{\omega}(X)\hat{\omega}(Y) = 2\hat{e}_2 + 2\hat{e}_2$ by the count in (1); since $\{t, u\} \mapsto \{v, z\}$ permutes the six pairs and $\hat{A}_{tu} \geq \min_{i \neq j} A_{ij}$, $\langle \nu, E_{\text{tot}} \rangle \geq 2 \min_{i \neq j} A_{ij} \hat{e}_2$. For F_A^w the first term of its pairing is ≥ 0 and the second is $\geq 2w(3) \min_{i \neq j} A_{ij} \hat{e}_2$. Equality: for $A_{ij} = 1$ ($i \neq j$) and $k = n - 2$, every k -partition has one block of three points or two blocks of two, so $E_{\text{tot}} \in \{2, 3\}$ and $|E_{\text{tot}} - \frac{5}{2}| = \frac{1}{2}$ ($K = 0$). $\square$

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